

ReDet: A Rotation-equivariant Detector for Aerial Object Detection

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Abstract

Recently, object detection in aerial images has gained much attention in computer vision. Different from objects in natural images, aerial objects are often distributed with arbitrary orientation. Therefore, the detector requires more parameters to encode the orientation information, which are often highly redundant and inefficient. Moreover, as ordinary CNNs do not explicitly model the orientation variation, large amounts of rotation augmented data is needed to train an accurate object detector. In this paper, we propose a Rotation-equivariant Detector (ReDet) to address these issues, which explicitly encodes rotation equivariance and rotation invariance. More precisely, we incorporate rotation-equivariant networks into the detector to extract rotation-equivariant features, which can accurately predict the orientation and lead to a huge reduction of model size. Based on the rotation-equivariant features, we also present Rotation-invariant RoI Align (RiRoI Align), which adaptively extracts rotation-invariant features from equivariant features according to the orientation of RoI. Extensive experiments on several challenging aerial image datasets DOTA-v1.0, DOTA-v1.5 and HRSC2016, show that our method can achieve state-of-the-art performance on the task of aerial object detection. Compared with previous best results, our ReDet gains 1.2, 3.5 and 2.6 mAP on DOTA-v1.0, DOTA-v1.5 and HRSC2016 respectively while reducing the number of parameters by 60% (313 Mb vs. 121 Mb). The code is available at: <https://github.com/csuhan/ReDet>.

1. Introduction

This paper studies the problem of object detection in aerial images, a recently-emerged challenging problem in

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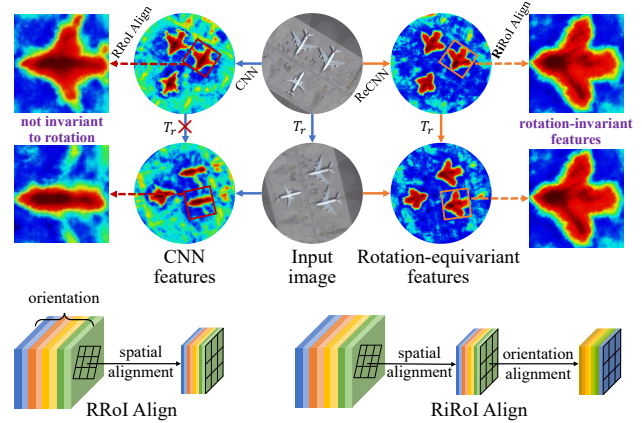


Figure 1. **Illustration of our method (top) and comparisons of RRoI warping (bottom).** CNN features are not equivariant to the rotation T_r , i.e., feeding a rotated image to CNNs is not the same as rotating feature maps of the original image. Therefore, the corresponding RoI features are not invariant to rotation. In contrast, our method adopts rotation-equivariant CNNs (ReCNN) to extract rotation-equivariant features. Let I and Φ be the input and ReCNN respectively, the equivariance of our method can be expressed as: $\Phi(T_r I) = T_r \Phi(I)$, i.e., applying a rotation T_r to the image I is the same as the rotation of features. Since we have obtained rotation-equivariant features, rotation-invariant features can be extracted by RRoI warping. While RRoI Align can only achieve rotation invariance in the spatial dimension, we present a novel Rotation-invariant RoI (RiRoI) Align to extract rotation-invariant features in both spatial and orientation dimensions.

computer vision [35]. Different from objects in nature images, objects in aerial images are often distributed with arbitrary orientation. To cope with these challenges, aerial object detection are usually formulated as an oriented object detection task by relying on Oriented Bounding Boxes (OBBs) representation instead of using Horizontal Bounding Boxes (HBBs) [7, 35, 38, 40].

Recently, many well-designed oriented object detectors have been proposed and reported promising results on challenging aerial image datasets [21, 35]. In order to achieve accurate object detection in unconstrained aerial images, most of them are devoted to extract rotation-invariant features [7, 10, 22, 37]. In practice, Rotated RoI (RRoI) warp-

ing (e.g., RRoI Pooling [22] and RRoI Align [7]) is the most commonly used method to extract rotation-invariant features, which can warp region features precisely according to the bounding boxes of RRoI in the 2D planar. However, RRoI warping with regular CNN features can not produce exactly rotation-invariant features. The rotation invariance is approximated by employing larger capacity networks and more training samples to model the rotation variation. As shown in Fig. 1, the regular CNNs are not equivariant to the rotation, *i.e.*, feeding a rotated image to CNNs is not the same as rotating feature maps of the original image. Therefore, region features warped from regular CNN feature maps are usually unstable and delicate as the orientation changes.

Some recently proposed methods [5, 13, 33] extend CNNs to larger groups and achieve rotation equivariance¹ with group convolutions [5]. Feature maps of these methods have additional orientation channels recording features from different orientations. However, directly applying the ordinary RRoI warping to rotation-equivariant features is unable to produce rotation-invariant features, as it can only warp region features in the 2D planar, *i.e.*, the spatial dimension, while the orientation channels are still misaligned. To extract completely rotation-invariant features, we also need to adjust the orientation dimension of feature maps according to the orientation of RRoI.

In this paper, we propose a Rotation-equivariant Detector (ReDet) to extract completely rotation-invariant features from rotation-equivariant features. As shown in Fig. 1, our method consists of two parts: rotation-equivariant feature extraction and rotation-invariant feature extraction. Firstly, we incorporate rotation-equivariant networks into the backbone to produce rotation-equivariant features, which can accurately predict the orientation and reduce the complexity of modeling orientation variations. Since directly apply the RRoI warping still cannot extract rotation-invariant features from the rotation-equivariant features, we propose a novel Rotation-invariant RoI Align (RiRoI Align). It can warp region features according to the bounding boxes of RRoI in the spatial dimension and align features in the orientation dimension by circularly switching orientation channels and feature interpolation. Finally, the combination of rotation-equivariant backbone and RiRoI Align forms our ReDet to extract completely rotation-invariant features for accurate aerial object detection.

Extensive experiments performed on the challenging aerial image datasets DOTA [35] and HRSC2016 [21] demonstrate the effectiveness of our method. We summary our contributions as: (1) We propose a Rotation-equivariant Detector for high-quality aerial object detection, which encodes both rotation equivariance and rotation invariance. To

¹Equivariance is a property that applying transformations to the input produces transformations of the feature in a predictable way.

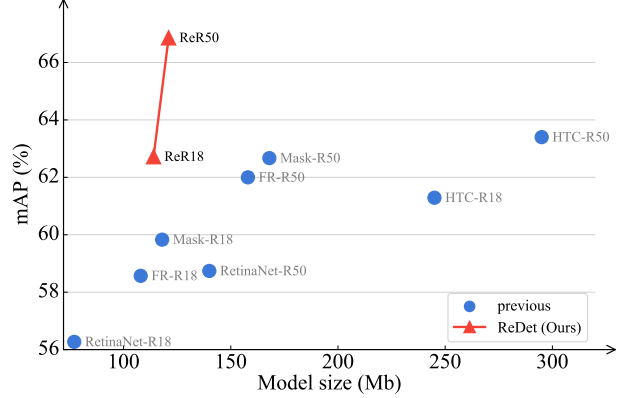


Figure 2. **Model size vs. accuracy (mAP) on DOTA-v1.5.** We evaluate RetinaNet OBB [18], Faster R-CNN OBB (FR) [27], Mask R-CNN (Mask) [11] and Hybrid Task Cascade (HTC) [2] with ResNet18 (R18) and ResNet50 (R50) backbones. Note all algorithms are our re-implemented version for DOTA, which is consistent with Tab. 7. Our ReDet is tested with ReResNet18 (ReR18) and ReResNet50 (ReR50) backbones. Compared with other methods with R18/R50 backbones, our ReDet with a ReR18 backbone achieves competitive performance. Using a deeper backbone (ReR50), our ReDet outperforms all methods by a large margin and achieves better model size vs. accuracy trade-off.

our best knowledge, it is the first time that rotation equivariance has been systematically introduced into oriented object detection. (2) We design a novel RiRoI Align to extract rotation-invariant features from rotation-equivariant features. Different from other RRoI warping methods, RiRoI Align produces completely rotation-invariant features in both spatial and orientation dimensions. (3) Our method achieves the state-of-the-art **80.10**, **76.80** and **90.46** mAP on DOTA-v1.0, DOTA-v1.5 and HRSC2016, respectively. Compared with previous best results, our method gains **1.2**, **3.5** and **2.6** mAP improvements. Compared with the baseline, our method shows consistent and substantial improvements and reduces the number of parameters by **60%** (313 Mb vs. 121 Mb). Moreover, our method achieves better model size vs. accuracy trade-off (shown in Fig. 2).

2. Related Works

2.1. Oriented Object Detection

Unlike most general object detectors [8, 9, 18, 20, 26, 27, 44] that use HBBs, oriented object detectors locate and classify objects with OBBs, which provide more accurate orientation information of objects. This is essential for detecting aerial objects with large aspect ratio, arbitrary orientation and dense distribution. With the development of general object detection, many well-designed methods have been proposed for oriented object detection [1, 7, 24, 35, 38, 40, 42], showing promising performance on challenging datasets [21, 35]. To detect objects with arbitrary orientation, some methods [1, 22, 43] adopt numerous rotated an-

chors with different angles, scales and aspect ratios for better regression while increasing the computation complexity. Ding *et al.* proposed RoI Transformer [7] to transform Horizontal RoIs (HRoIs) into RRoIs, which avoids a large number of anchors. Gliding vertex [36] and CenterMap [30] use quadrilateral and mask to accurately describe oriented objects, respectively. R³Det and S²A-Net align the feature between horizontal receptive fields and rotated anchors. DRN [24] detects oriented objects with dynamic feature selection and refinement. CSL [38] regards angular prediction as a classification task to avoid discontinuous boundaries problem. Recently, some CenterNet [44]-based methods [24, 31, 41] show their advantages in detecting small objects. The above methods are devoted to improving object representations or feature representations. While our method is dedicated to improving the feature representation throughout the network: from the backbone to the detection head. Specifically, our method produces rotation-equivariant features in the backbone, significantly reducing the complexity in modeling orientation variations. In the detection head, the RiRoI Align extracts completely rotation-invariant features for robust object localization.

2.2. Rotation-equivariant Networks

Cohen *et al.* first proposed group convolutions [5] to incorporate 4-fold rotation equivariance into CNNs. HexaConv [13] extends group convolutions to 6-fold rotation equivariance over hexagonal lattices. To achieve rotation equivariance on more orientations, some methods [23, 45] resampling filters by interpolation, while other methods [32, 33, 34] use harmonics as filters to produce equivariant features in the continuous domain. The above methods gradually extend rotation equivariance to larger groups and achieve promising results on the classification task, while our method incorporates rotation-equivariant networks into the object detector, showing significant improvements on the detection task. To our best knowledge, this is the first time that rotation equivariance has been systematically applied to oriented object detection.

2.3. Rotation-invariant Object Detection

The rotation-invariant feature is important for detecting arbitrary oriented objects. However, CNNs show poor performance in modeling rotation variations, which means that more parameters are needed to encode the orientation information. STN [14] and DCN [6] explicitly model the rotation within the network and have been widely applied to oriented object detection [7, 28, 29]. Cheng *et al.* [4] proposed a rotation-invariant layer that imposes an explicit regularization constraint to the objective. Though the above methods can achieve approximated rotation invariance in the image-level, large amounts of training samples and parameters are needed. Besides, object detection requires instance-level

rotation-invariant features. Therefore, some methods [7, 22] extend RoI warping [8] to RRoI warping, *e.g.*, RoI Transformer [7] learns to transform HRoIs to RRoIs and warps region features with a rotated position sensitive RoI Align. However, the regular CNNs are not rotation-equivariant. Therefore, even through the RRoI Align, we still cannot extract rotation-invariant features, as shown in Fig. 1. Different from the aforementioned methods, our method proposes Rotation-invariant RoI Align (RiRoI Align) to extract rotation-invariant features from rotation-equivariant features. Specifically, we incorporate rotation-equivariant networks into the backbone to produce rotation-equivariant features, then the RiRoI Align extracts completely rotation-invariant features from rotation-equivariant features in both spatial and orientation dimensions.

3. Preliminaries

Equivariance is a property that applying transformations to the input produces transformations of the feature in a predictable way. Formally, give a transformation group G and a function $\Phi : X \rightarrow Y$, equivariance can be expressed as:

$$\Phi[T_g^X(x)] = T_g^Y[\Phi(x)] \quad \forall (x, g) \in (X, G), \quad (1)$$

where T_g indicates a group action in the corresponding space. Especially when T_g^Y is identical for all T_g^X , equivariance becomes invariance.

In common, CNNs are known to be translation equivariant. Let T_t denotes an action of the translation group $(\mathbb{R}^2, +)$, and apply it to K -dimension feature maps $f : \mathbb{Z}^2 \rightarrow \mathbb{R}^K$, translation equivariance can be expressed as:

$$[[T_t f] * \psi](x) = [T_t [f * \psi]](x), \quad (2)$$

where $\psi : \mathbb{Z}^2 \rightarrow \mathbb{R}^K$ indicates the convolution filter and $*$ is the convolution operation. Recently proposed methods [5, 13, 33] extend CNNs to large groups, achieving both translation and rotation equivariance. Let H denotes a rotation group, *e.g.*, the cyclic group C_N containing discrete rotations by angles multiple of $\frac{2\pi}{N}$. We can define the group G as the semidirect product of the translation group $(\mathbb{R}^2, +)$ and the rotation group H , *i.e.*, $G \cong (\mathbb{R}^2, +) \rtimes H$. By replacing $x \in (\mathbb{R}^2, +)$ with $g \in G$ in Eq. 2, the rotation-equivariant convolution can be defined as:

$$[[T_g f] * \psi](g) = [T_g [f * \psi]](g). \quad (3)$$

Rotation-equivariant Networks. The regular CNNs consists of a series of convolution layers and enjoy the translation weight sharing. Similarly, rotation-equivariant networks are a stack of rotation-equivariant layers with a higher degree of weight sharing, *i.e.*, both translation and rotation. Formally, let $\Phi = \{L_i | i \in \{1, 2, \dots, M\}\}$ denotes a network with M rotation-equivariant layers under